

Akshay Kumar

AI & Backend Engineer - agentic systems, LLM pipelines, async services

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PROFILE

Software Engineer building agentic systems, LLM pipelines, async services and API integrations from architecture through deployment. Recent work spans autonomous publishing, AI video processing, market intelligence and automation using Python, FastAPI, TypeScript, Docker, Redis and PostgreSQL.

SKILLS

Languages: Python, TypeScript, JavaScript, Rust, SQL **Backend:** FastAPI, Node.js, REST APIs, OAuth 2.0, Docker, PostgreSQL, Redis, Celery, WebSockets **Frontend:** React, Next.js, Astro, React Native, Tailwind CSS **AI/Data:** LLM APIs, Whisper, FFmpeg, Computer Vision, NLP, ChromaDB, TimescaleDB

EXPERIENCE - Freelance Software Engineer (2024 - Present)

- Delivered backend and AI-assisted automation systems for private clients, owning requirements, architecture, integrations and deployment.
- Built multi-tenant SaaS analytics product for keyword research, account analytics, competitor tracking and comment mining with tiered plans.
- Architected Facebook content pipeline for 10+ pages: Telegram ingestion, niche detection, scheduling, encrypted credentials, Graph API publishing.

SELECTED PROJECTS (all built & open-source; first two live in production)

- **BrowserPixels** - browserpixels.com - privacy-first in-browser image toolkit (compress, convert, resize, GIF, EXIF strip, zero uploads). *Astro, React, TypeScript, Cloudflare Workers*. github.com/aksaayyy/browserpixels
- **Woof & Wag** - woofnwagg.com - production dog-care platform, 80+ vet-reviewed articles, Sanity auto-upload pipeline, SEO. *Next.js, TypeScript, Sanity CMS*. github.com/aksaayyy/DogWebsite
- **ClipStream** - YouTube-to-clips microservice: download, Whisper transcription, engagement scoring, vertical rendering. *FastAPI, Whisper, FFmpeg, Docker*. github.com/aksaayyy/ClipStream
- **SyncSphere** - trip photo-sharing with QR join, RBAC, activity tracking, CV clustering. *FastAPI, SQLAlchemy, React Native, CV*. github.com/aksaayyy/Syncsphere
- **Article Forge** - autonomous AI publishing: topic planning, generation, media, linking, QA, Sanity publish. *Python, AI, Sanity CMS*. github.com/aksaayyy/article-forge
- **Vext-ai** - extracts code/config from video tutorials via multi-model LLMs, Dockerfiles/Terraform/CLI output. *Next.js, TypeScript, Groq/Claude/OpenRouter*. github.com/aksaayyy/Vext-ai
- **Facebook Auto Bot** - scheduled multi-page posting backend, OAuth 2.0, rate limiting, encrypted tokens. *Python, Graph API*. github.com/aksaayyy/Facebook_Auto_Bot

EDUCATION

B.E. - Computer Science & Engineering (Information Security) - Chandigarh University, 2025